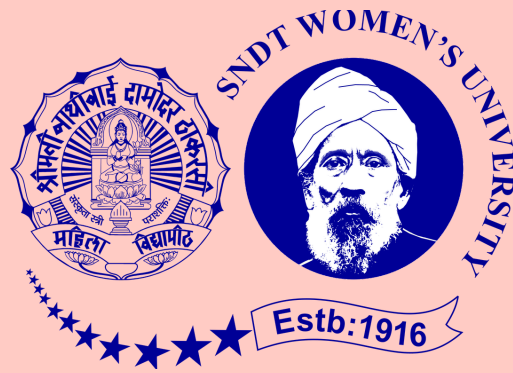


SCALABLE SPATIOTEMPORAL ESTIMATION OF PARTICULATE MATTER VIA MULTI SOURCE DATA FUSION AND DEEP LEARNING - AIRSENSE

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Abstract

Ground monitoring stations provide accurate data but lack spatial coverage, while satellite-derived Aerosol Optical Depth (AOD) offers wide-area representation without directly estimating surface-level pollution. This study proposes a hybrid machine learning framework integrating INSAT AOD imagery, CPCB ground observations, and meteorological parameters for improved PM estimation. By combining a meteorology-based ground model with image-derived corrections, the approach generates spatially continuous and more robust surface-level PM predictions suitable for scalable urban air quality monitoring.

Objective

- Develop a hybrid PM prediction model using satellite AOD and weather data.
- Enhance ground-level estimates using image-based correction modeling.
- Generate spatial pollution maps for Mumbai.
- Build a real-time air quality monitoring and prediction system.

Data Source

1. Ground Observations (MPCB)

- Daily Air Quality Index (AQI) and particulate matter data
- Monitoring Station: Bandra Kurla Complex (BKC), Mumbai
- Study Period: January 2023 – October 2025

2. Satellite Data (Aerosol Optical Depth)

- INSAT-3DR Level-2 (L2G) AOD products
- 7–8 temporal captures per calendar date
- Study Period: January 2023 – October 2025

Methods

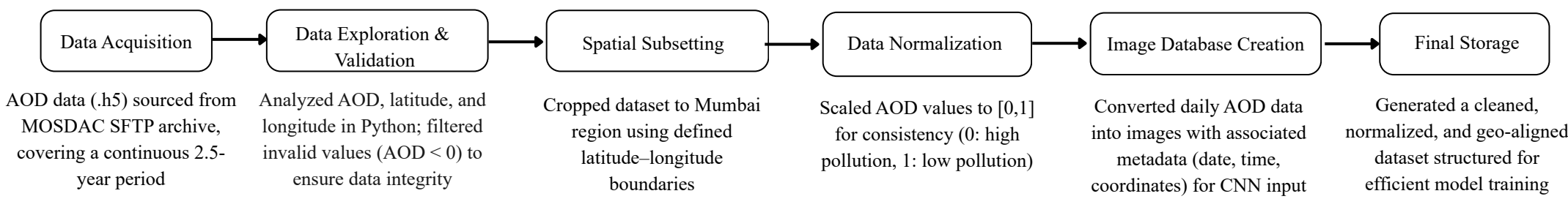


Fig.3 Image Preprocessing pipeline

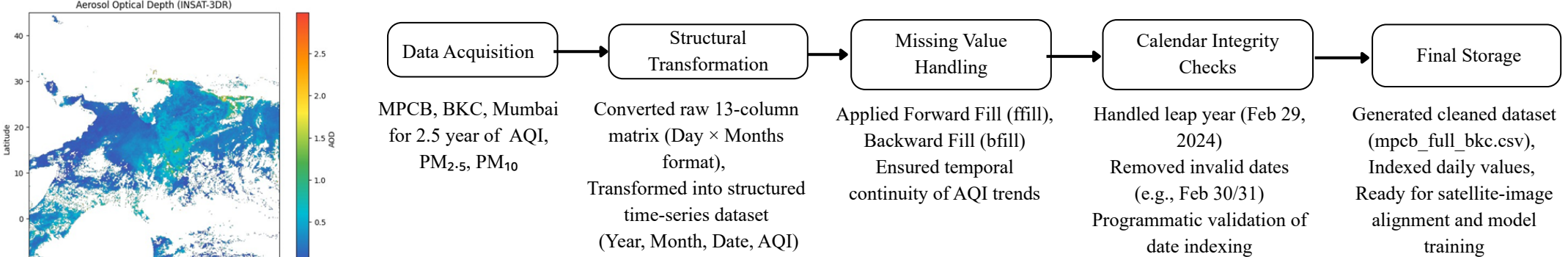


Fig.4 Ground Data Preparation Pipeline

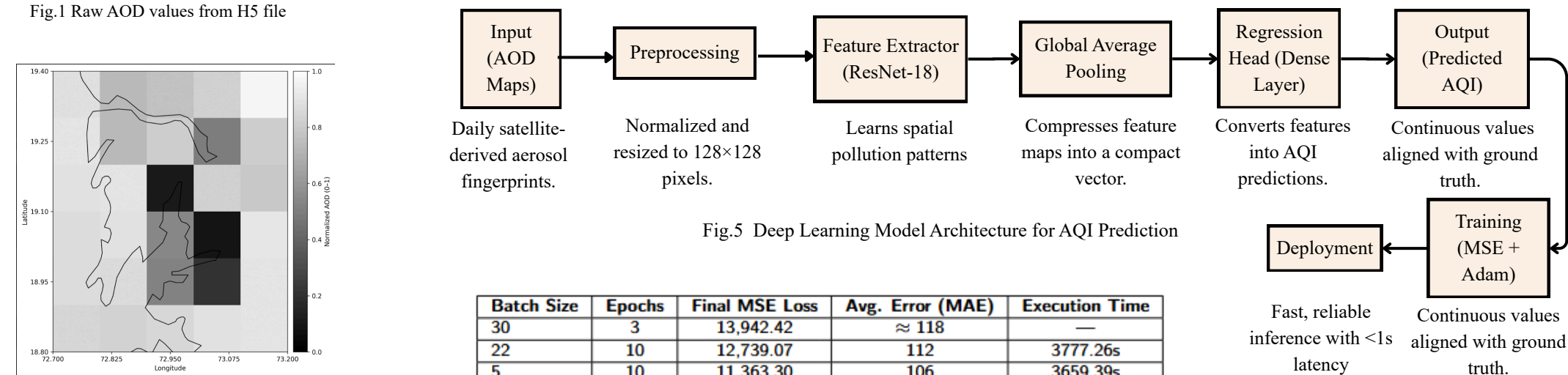


Fig.5 Deep Learning Model Architecture for AQI Prediction

Batch Size	Epochs	Final MSE Loss	Avg. Error (MAE)	Execution Time
30	3	13,942.42	≈ 118	—
22	10	12,739.07	112	3777.26s
5	10	11,363.30	106	3659.39s
50	7	13,423.70	10*	19,258.09s

Table.1 Hyperparameter Tuning Trials (ResNet-18 + Regression Head)

Publications

